# The Meridian Study — Participant Experience

**Post-Session Usability Evaluation & Think-Aloud Debrief**

Report A | Identifying header fields removed from this public copy.

**Instructions:** Thank you for participating in this session. We are evaluating the ease of use, clarity, and interface design of the Meridian Subject Portal. Your honest impressions are essential to help us improve the system. There are no right or wrong answers, and your feedback evaluates the software, not your personal ability.

## Part 1: System Usability Scale (SUS)

*Please indicate your level of agreement with each statement regarding your interaction with the system today.*

| # | Statement | Strongly Disagree (1) | Disagree (2) | Neutral (3) | Agree (4) | Strongly Agree (5) |
|---|---|---|---|---|---|---|
| 1 | I would like to use this system frequently for such studies. | 1 | 2 | 3 | 4 | 5 |
| 2 | I found the system unnecessarily complex or confusing. | 1 | 2 | 3 | 4 | 5 |
| 3 | I thought the system was easy to use. | 1 | 2 | 3 | 4 | 5 |
| 4 | I think I would need the support of an instructor to use it. | 1 | 2 | 3 | 4 | 5 |
| 5 | I found the various functions and tasks were well integrated. | 1 | 2 | 3 | 4 | 5 |
| 6 | I thought there was too much inconsistency across the tasks. | 1 | 2 | 3 | 4 | 5 |
| 7 | I imagine that most people would learn to use this very quickly. | 1 | 2 | 3 | 4 | 5 |
| 8 | I found the system cumbersome or frustrating to use. | 1 | 2 | 3 | 4 | 5 |
| 9 | I felt very confident using the interface. | 1 | 2 | 3 | 4 | 5 |
| 10 | I needed to learn a lot before I could get started with the tasks. | 1 | 2 | 3 | 4 | 5 |

## Part 2: Module & Task Assessment

*Rate the clarity and operational ease of each stage you encountered during the protocol:*

| Protocol Stage / Component | Clear & Seamless | Moderate / Acceptable | Unclear / High Effort |
|---|---|---|---|
| **Session Entry:** Connecting via session code or participant ID | ☐ | ☐ | ☐ |
| **Stimulus Playback:** Watching the emotion elicitation video clips | ☐ | ☐ | ☐ |
| **Self-Report Ratings:** Submitting Valence, Arousal, and Dominance | ☐ | ☐ | ☐ |
| **Stroop Task:** Matching ink colors against printed color words | ☐ | ☐ | ☐ |
| **N-Back Task:** Tracking and matching spatial positions from $N$ steps ago | ☐ | ☐ | ☐ |
| **Physiological Sweep:** Breathing guides (orb pacer, breath-hold) | ☐ | ☐ | ☐ |
| **System Communication:** Instructor announcements and stage transition notes | ☐ | ☐ | ☐ |

## Part 3: Retrospective Think-Aloud Debriefing

*These questions explore specific verbalized observations and cognitive moments from the think-aloud session.*

**1. Positive Aspects & Clarity:**
Which task or interface element felt the most intuitive, responsive, or straightforward to navigate?

Colouring task

**2. Cognitive Load & Friction Points:**
Were there any moments during the session where you felt uncertain, paused unexpectedly, or experienced confusion regarding what action was expected of you?

N-back test

**3. Safety & Autonomous Navigation Awareness:**
During live execution, safety and navigational controls are visible at the bottom of the portal (e.g., *"I need to stop"*, *"Previous task"*, *"Next task"*). How clear were these options to you?

- ☐ I noticed them clearly and felt comfortable using them if necessary.
- ☑ I observed them, but was uncertain when or whether it was appropriate to interact with them.
- ☐ I did not notice them during the session.

**4. Constructive Recommendations:**
If you could modify one visual element, prompt, or instruction in this portal to make the workflow smoother, what would you suggest?

font size

**Thank you for contributing your insights to the Meridian Study.**

## Researcher Reference: SUS Scoring & Think-Aloud Administration

### Standard Brooke (1996) SUS Calculation Algorithm

Each participant's response $S_i \in \{1, 2, 3, 4, 5\}$ across the 10 items is converted to a normalized item contribution $X_i$:

$$X_i = \begin{cases} S_i - 1 & \text{for odd items } (i \in \{1, 3, 5, 7, 9\}) \\ 5 - S_i & \text{for even items } (i \in \{2, 4, 6, 8, 10\}) \end{cases} \quad \Longrightarrow \quad \text{SUS Score} = 2.5 \times \sum_{i=1}^{10} X_i \quad \in [0, 100]$$

| Item | Brooke (1996) Construct | Standardized Plain Phrasing (All Cohorts) | Item Weight |
|---|---|---|---|
| Q1 | System adoption | I would like to use this system frequently for such studies. | $S_1 - 1$ |
| Q2 | Complexity | I found the system unnecessarily complex or confusing. | $5 - S_2$ |
| Q3 | Ease of use | I thought the system was easy to use. | $S_3 - 1$ |
| Q4 | Support dependence | I think I would need the support of an instructor to use it. | $5 - S_4$ |
| Q5 | System integration | I found the various functions and tasks were well integrated. | $S_5 - 1$ |
| Q6 | Inconsistency | I thought there was too much inconsistency across the tasks. | $5 - S_6$ |
| Q7 | Learnability | I imagine that most people would learn to use this very quickly. | $S_7 - 1$ |
| Q8 | Cumbersomeness | I found the system cumbersome or frustrating to use. | $5 - S_8$ |
| Q9 | User confidence | I felt very confident using the interface. | $S_9 - 1$ |
| Q10 | Knowledge prerequisite | I needed to learn a lot before I could get started with the tasks. | $5 - S_{10}$ |

### Empirical Benchmarks (Sauro & Lewis, 2016; Bangor et al., 2008)

- $\geq 80.3$ **(Grade A / Excellent)**: Exceptional usability; participant operates with minimal cognitive friction.
- $68.0 - 80.2$ **(Grade B/C / Good)**: Standard industry average ($\sim 68$). Functional for controlled experimental runs.
- $51.0 - 67.9$ **(Grade D / Marginal)**: Noticeable friction points; potential confounding of physiological data by interface struggle.
- $< 51.0$ **(Grade F / Poor)**: Severe usability deficits requiring protocol redesign prior to data collection.

### Controlled Think-Aloud Administration Across Diverse Age Cohorts

1. **Concurrent Verbalization:** Instruct the participant prior to starting: *"Please speak your thoughts aloud continuously as you read, react, click, or wait. Describe what you are looking at and what you expect to happen."*
2. **Neutral Facilitator Probing:** If a participant lapses into silence for $> 10$ seconds (common during cognitively demanding tasks like N-back or Stroop), deliver strictly non-evaluative prompts: *"What are you looking at right now?"* or *"Please keep talking aloud."* Avoid biasing queries like *"Is this difficult?"* or *"Did you mean to click that?"*.
3. **Post-Session Timing:** Administer Parts 1–3 immediately after the subject portal reaches the terminal screen (`Session complete / Thank you`), before discussing experimental objectives.